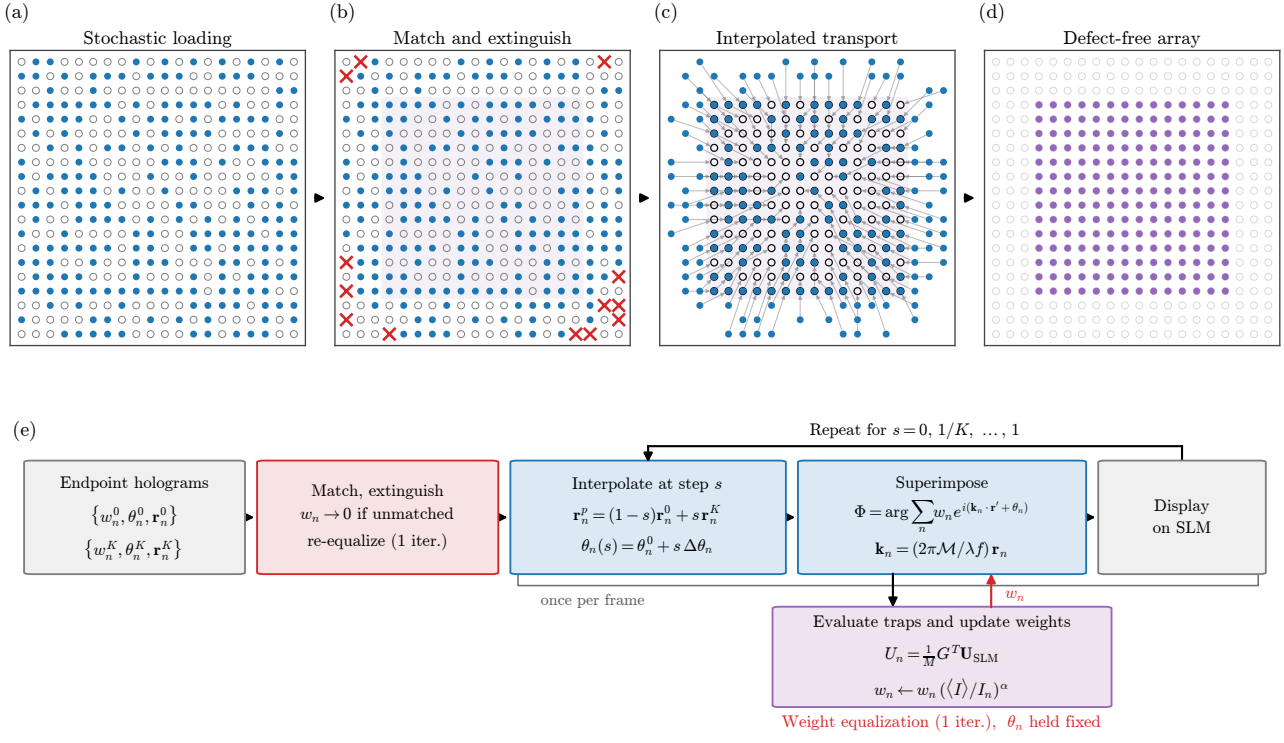
(a)
Stochastic loading
(b)
Match and extinguish
(c)
Interpolated transport
(d)
Defect-free array
(e)
Repeat for $s = 0,\ 1/K,\ \ldots,\ 1$
Endpoint holograms
$\{w_n^0, \theta_n^0, \mathbf{r}_n^0\}$
$\{w_n^K, \theta_n^K, \mathbf{r}_n^K\}$
Match, extinguish
$w_n \to 0$ if unmatched
re-equalize (1 iter.)
Interpolate at step s
$\mathbf{r}_n^p = (1-s)\mathbf{r}_n^0 + s\,\mathbf{r}_n^K$
$\theta_n(s) = \theta_n^0 + s\,\Delta\theta_n$
Superimpose
$\Phi = \arg\sum_n w_n e^{i(\mathbf{k}_n \cdot \mathbf{r}' + \theta_n)}$
$\mathbf{k}_n = (2\pi\mathcal{M}/\lambda f)\,\mathbf{r}_n$
Display
on SLM
once per frame
w_n
Evaluate traps and update weights
$U_n = \frac{1}{M} G^T \mathbf{U}_{\mathrm{SLM}}$
$w_n \leftarrow w_n \left(\langle I \rangle / I_n\right)^{\alpha}$
Weight equalization (1 iter.), θ_n held fixed